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Expected value and probability tools

Level 3 Senior · Combinatorics

Compute expected values with linearity and indicators, solve repeated processes with states, and handle conditional probability and Bayes' rule.

By the end you can: You will be able to find expected values quickly by writing a count as a sum of indicators; You will be able to set up and solve state equations for probabilities and expected waiting times; You will be able to use conditional probability and Bayes' rule without falling into the classic traps.

Lesson

Twelve people at a party throw their hats into a pile, and each takes one back at random. On average, how many of them get their own hat? The number of people with their own hat has a complicated distribution: working out the probability that exactly three do needs the inclusion-exclusion of the last unit. Yet the average is exactly 1, and here is the whole argument. Each person gets their own hat with probability $\frac{1}{12}$, there are 12 people, so the expected number is $12 \cdot \frac{1}{12} = 1$. The same argument gives 1 for any number of people.

That argument used a tool called *linearity of expectation*, and this unit is about it and two others: *states*, for processes that go on until something happens, and *conditional probability*, for updating when you learn something. They appear on the SMC and Senior Kangaroo and all over the AMC 12 and AIME.

Expected value

If a random quantity X takes values $x_1, x_2, \ldots$ with probabilities $p_1, p_2, \ldots$, its *expected value* is

$$E[X] = \sum_i x_i p_i.$$

Think of it as the long-run average if you repeated the experiment many times. For one roll of a fair die, $E[X] = \frac{1+2+3+4+5+6}{6} = \frac{7}{2}$. Notice that $\frac{7}{2}$ is not a possible roll; the expected value is an average, not a prediction.

Linearity and indicators

> **Linearity of expectation.** For any random quantities X and Y defined on the same experiment, and any constant c,
>
> $$E[X+Y] = E[X] + E[Y] \quad \text{and} \quad E[cX] = c\,E[X].$$
>
> This holds whether or not X and Y are independent.

The reason is short. Add up over the individual outcomes ω of the experiment: $E[X] = \sum_\omega X(\omega)P(\omega)$, so

$$E[X+Y] = \sum_\omega \big(X(\omega)+Y(\omega)\big)P(\omega) = \sum_\omega X(\omega)P(\omega) + \sum_\omega Y(\omega)P(\omega) = E[X] + E[Y].$$

Nothing in this used independence.

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The way to use it is through *indicators*. For an event A, let I_A be 1 if A happens and 0 if not. Then $E[I_A] = 1 \cdot P(A) + 0 \cdot (1 - P(A)) = P(A)$. If you can write a count X as a sum of indicators, $X = I_{A_1} + \cdots + I_{A_m}$, then

$$E[X] = P(A_1) + \cdots + P(A_m).$$

The events can be wildly dependent and it does not matter. That is the whole trick, and it is far more powerful than it looks.

Example 1. A fair die is rolled 12 times. A *rise* is a roll that is larger than the roll just before it. What is the expected number of rises?

Solution.
Finding the distribution of the number of rises is hopeless: it can be anything from 0 to 11, with very uneven probabilities. So do not try. For each i from 1 to 11, let I_i be 1 if roll $i+1$ is larger than roll i. The number of rises is $I_1 + \cdots + I_{11}$.
Each I_i looks at just two rolls. Of the 36 equally likely ordered pairs, 6 are ties, and by symmetry the other 30 split evenly between "second larger" and "first larger". So $P(I_i = 1) = \frac{15}{36} = \frac{5}{12}$, and the expected number of rises is $11 \cdot \frac{5}{12} = \frac{55}{12}$.
Note that I_1 and I_2 are not independent, since both involve roll 2. Linearity does not care.

Example 2. Six friends each choose a day of the week to go to the gym, independently and uniformly at random. What is the expected number of days on which at least one of them goes?

Solution.
The natural first idea is to put an indicator on each friend, but "friend j adds a new day" depends on all the earlier friends, so those indicators are awkward. The trick is to choose indicators that are easy. Put one on each *day* instead: let J_d be 1 if at least one friend goes on day d.
The complement is easy: nobody goes on Monday with probability $\left(\frac{6}{7}\right)^6$, since each friend independently avoids it. So $P(J_d = 1) = 1 - \left(\frac{6}{7}\right)^6$, and the expected number of days used is

$$7\left(1 - \left(\tfrac{6}{7}\right)^6\right) = 7 - \frac{6^6}{7^5} = \frac{70993}{16807} \approx 4.22.$$

Sanity check: the answer lies between 1 and 6, and it is noticeably less than 6 because friends often clash.

States and recursion

Many questions describe a process that runs until something happens. The tool is to condition on the first step. If $B_1, \ldots, B_m$ are the possible outcomes of the first step, then

$$P(\text{win}) = \sum_i P(B_i)\, P(\text{win} \mid B_i),$$

and the same holds with expected values in place of probabilities. In a process, "what happens after B_i" is often the same question started from a new position, or *state*, which gives equations.

The state method. Choose states that record everything about the past that matters for the future. Let e_s be the expected number of further steps from state s, with $e_s = 0$ at the end. Then

$$e_s = 1 + \sum_t P(s \to t)\, e_t.$$

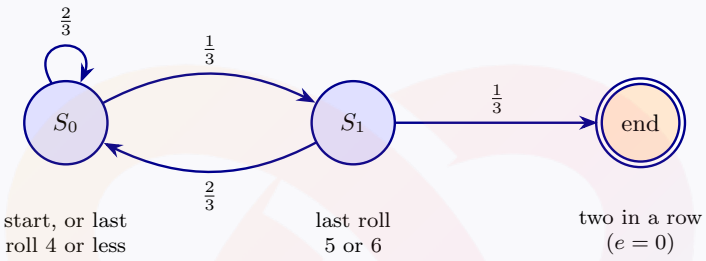
For the probability p_s of finishing in a winning state, $p_s = \sum_t P(s \to t)\, p_t$, with $p = 1$ at a win and $p = 0$ at a loss.

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The 1 is the step you are about to take. The simplest case is waiting for an event of probability p on each independent trial: one state, so $e = 1 + (1-p)e$, and $e = \frac{1}{p}$. (These equations are valid once you know the expected values are finite. That is true in every process at this level where the end can be reached from every state, so we will not fuss about it.)

Example 3. A fair die is rolled until two consecutive rolls are both 5 or more. What is the expected number of rolls?

Solution.
A tempting wrong answer: each pair of rolls succeeds with probability $\frac{1}{3} \cdot \frac{1}{3} = \frac{1}{9}$, so expect about 9 pairs. But consecutive pairs overlap, so they are not independent trials, and that reasoning has no basis. We need states.
What matters about the past? Only whether the last roll was a 5 or 6. So use S_0 (at the start, or the last roll was 4 or less) and S_1 (the last roll was 5 or 6). Each roll is 5 or more with probability $\frac{1}{3}$, which gives the diagram below.



Reading off the arrows,

$$e_0 = 1 + \tfrac{1}{3}e_1 + \tfrac{2}{3}e_0, \qquad e_1 = 1 + \tfrac{1}{3} \cdot 0 + \tfrac{2}{3}e_0.$$

The first gives $e_0 = 3 + e_1$. Substituting the second, $e_0 = 4 + \frac{2}{3}e_0$, so $e_0 = 12$. The answer is 12 rolls, quite a lot more than the naive guess.

Conditional probability

The probability of A given B is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

In words: throw away every outcome outside B, and see what fraction of what is left lies in A. Combining this with the total probability formula gives *Bayes' rule*,

$$P(B \mid A) = \frac{P(A \mid B)\,P(B)}{P(A)},$$

where $P(A)$ is found by splitting into cases. The wording of the condition matters. Roll two dice. Given that *the first* shows a six, both are sixes with probability $\frac{1}{6}$. Given that *at least one* shows a six, there are 11 equally likely outcomes left, so the probability is only $\frac{1}{11}$.

Example 4. A box holds three coins: a fair coin, a coin with heads on both sides, and a bent coin that lands heads with probability $\frac{1}{4}$. A coin is taken at random and tossed twice, landing heads both times. What is the probability that a third toss of the same coin is also heads?

Solution.
A tempting wrong answer: a random coin lands heads with probability $\frac{1}{3}\left(\frac{1}{2} + 1 + \frac{1}{4}\right) = \frac{7}{12}$, so the answer is $\frac{7}{12}$. That treats the tosses as independent, but they are linked through the coin:

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two heads is evidence about which coin we hold. So first update our beliefs about the coin. The probabilities of two heads with each coin are $\frac{1}{4}$, 1 and $\frac{1}{16}$. Each coin was chosen with probability $\frac{1}{3}$, so $P(\text{HH}) = \frac{1}{3}\left(\frac{1}{4} + 1 + \frac{1}{16}\right) = \frac{7}{16}$, and by Bayes' rule the coins now have probabilities $\frac{4}{21}$, $\frac{16}{21}$ and $\frac{1}{21}$ (each is its $\frac{1}{3} \cdot P(\text{HH} \mid \text{coin})$ divided by $\frac{7}{16}$; check that they add to 1).
Now condition on the coin for the third toss:

$$\frac{4}{21} \cdot \frac{1}{2} + \frac{16}{21} \cdot 1 + \frac{1}{21} \cdot \frac{1}{4} = \frac{8 + 64 + 1}{84} = \frac{73}{84}.$$

This is much more than $\frac{7}{12}$, which makes sense: after two heads, the double-headed coin is the likely suspect.

Common traps.

- Linearity is for sums. In general $E[XY] \neq E[X]E[Y]$ (this needs independence), and $E[X^2] \neq E[X]^2$.
- An expected value need not be a possible value, and it is not the most likely value.
- States must remember everything that matters. Waiting for a pattern such as HTH, a failure does not always send you back to the start.
- Forgetting the $+1$ for the step being taken, or forgetting that the end state has $e = 0$.
- "Given at least one" and "given the first" are different conditions with different answers.
- Drawing without replacement makes draws dependent, but each single draw still has the same probability, so indicators still work.

How to spot this in a contest

The words "expected", "average" or "on average" with a count (how many fixed points, runs, pairs, records, distinct values, regions) almost always mean indicators: decide what each indicator should look at, find one probability, and multiply. If you find yourself computing a full distribution just to average it, stop and look for indicators.

If a process repeats until something happens (a game, a walk, a frog on lily pads, tossing until a pattern appears), define states and condition on the next step; look for symmetry to cut down the number of unknowns. If a question says "given that" or "if it is known that", write down the restricted sample space or use Bayes' rule, and read the condition twice to see exactly what is known.

Practice

Try each problem before turning to the hints and solutions at the back.

1. ⋆ A bag contains 5 red and 3 blue counters. Three counters are drawn at random without replacement. What is the expected number of red counters drawn?

2. ⋆⋆ Four boys and four girls sit down in a random order around a circular table with eight seats. What is the expected number of pairs of neighbours consisting of one boy and one girl?

3. ⋆⋆ In a school, 60% of the students study French. Of the students who study French, 30% also study German. Of the students who do not study French, 50% study German. A student chosen at random studies German. What is the probability that this student studies French?

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4. ⋆⋆⋆ A bug starts at a vertex A of a cube. Every minute it walks along an edge to one of the three neighbouring vertices, chosen at random with equal probabilities. What is the expected number of minutes until the bug first returns to A?

5. ⋆⋆⋆ The numbers $1, 2, \ldots, 10$ are arranged in a row in a random order. A number is called a *dip* if it is smaller than each of its neighbours. (The two numbers at the ends have only one neighbour each.) What is the expected number of dips?

6. ⋆⋆⋆⋆ A token starts at 0 on the number line. On each move it goes up by 1 with probability $\frac{2}{3}$ and down by 1 with probability $\frac{1}{3}$, independently of earlier moves. It stops as soon as it reaches 3 or -3. What is the expected number of moves?

Contest ladder

Real past problems that use this idea, easiest first. Open them on the official paper or the AoPS wiki, or online at kevinvariant08.github.io.

1. UKMT Senior Kangaroo 2021 Question 11 ⋆
2. UKMT Senior Kangaroo 2023 Question 15 ⋆⋆
3. AMC 10A 2017 Problem 18 ⋆⋆
4. UKMT Senior Kangaroo 2018 Question 5 ⋆⋆
5. AMC 10B Fall 2021 Problem 16 ⋆⋆
6. AMC 12B Fall 2021 Problem 23 ⋆⋆⋆
7. UKMT Senior Kangaroo 2024 Question 18 ⋆⋆⋆
8. AIME I 2019 Problem 5 ⋆⋆⋆
9. AMC 12B 2014 Problem 22 ⋆⋆⋆
10. AIME I 2023 Problem 6 ⋆⋆⋆
11. AIME II 2017 Problem 9 ⋆⋆⋆⋆
12. AIME I 2025 Problem 13 ⋆⋆⋆⋆

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Hints

1. What is the probability that the first counter drawn is red? What about the second, if you know nothing about the first?
2. There are eight pairs of neighbouring seats. Find the probability that one particular pair is mixed.
3. Work out what fraction of all students study German, and what fraction study both.
4. Group the vertices by their distance from A along the edges. For each distance, how many of the three neighbours are closer and how many further away?
5. Use an indicator for each position. For a middle position, only the order of three numbers matters.
6. Let e_k be the expected number of further moves from position k, so $e_3 = e_{-3} = 0$, and condition on the next move. The differences $e_{k-1} - e_k$ satisfy a simple recurrence.

Solutions

Problem 1

Answer: $\frac{15}{8}$

Let I_k be 1 if the k-th counter drawn is red. By symmetry, any particular draw is equally likely to be any of the 8 counters, so $P(I_k = 1) = \frac{5}{8}$ for each k, even though the draws are dependent. The number of reds is $I_1 + I_2 + I_3$, so by linearity its expected value is $3 \cdot \frac{5}{8} = \frac{15}{8}$.

Problem 2

Answer: $\frac{32}{7}$

There are 8 pairs of neighbouring seats. For one particular pair, the first seat is equally likely to hold any of the 8 people and the second any of the other 7. The pair is mixed if the first is a boy and the second a girl, with probability $\frac{4}{8} \cdot \frac{4}{7} = \frac{2}{7}$, or the other way round, also $\frac{2}{7}$. So each pair is mixed with probability $\frac{4}{7}$, and by linearity the expected number of mixed pairs is $8 \cdot \frac{4}{7} = \frac{32}{7}$.

Problem 3

Answer: $\frac{9}{19}$

The fraction of all students who study both languages is $0.6 \times 0.3 = 0.18$, and the fraction who study German but not French is $0.4 \times 0.5 = 0.20$. So 0.38 of the students study German, and the required probability is

$$P(\text{French} \mid \text{German}) = \frac{0.18}{0.38} = \frac{9}{19}.$$

Problem 4

Answer: 8

Group the vertices by their distance from A along edges: 3 vertices at distance 1, 3 at distance 2 and 1 at distance 3. A vertex at distance 1 has A and two vertices at distance 2 as neighbours. A vertex at distance 2 has two neighbours at distance 1 and one at distance 3. The vertex at distance 3 has all three neighbours at distance 2. By symmetry the expected time to reach A depends only

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on the distance, so let f_1, f_2 and f_3 be these expected times. Conditioning on the next move,

$$f_1 = 1 + \tfrac{2}{3}f_2, \qquad f_2 = 1 + \tfrac{2}{3}f_1 + \tfrac{1}{3}f_3, \qquad f_3 = 1 + f_2.$$

Substituting the third equation into the second gives $\frac{2}{3}f_2 = \frac{4}{3} + \frac{2}{3}f_1$, so $f_2 = 2 + f_1$. Then the first equation gives $f_1 = 1 + \frac{2}{3}(2 + f_1)$, so $\frac{1}{3}f_1 = \frac{7}{3}$ and $f_1 = 7$. The bug's first move always takes it to distance 1, so the expected return time is $1 + f_1 = 8$.

Problem 5

Answer: $\frac{11}{3}$

Let I_k be 1 if the number in position k is a dip. For an end position, the number there and its single neighbour are equally likely to be in either order, so $P(I_1 = 1) = P(I_{10} = 1) = \frac{1}{2}$. For a middle position k with $2 \le k \le 9$, the three numbers in positions $k-1$, k and $k+1$ are equally likely to appear in any of their 6 relative orders, and the middle one is the smallest of the three in 2 of them. So $P(I_k = 1) = \frac{1}{3}$.

By linearity the expected number of dips is $2 \cdot \frac{1}{2} + 8 \cdot \frac{1}{3} = 1 + \frac{8}{3} = \frac{11}{3}$.

Problem 6

Answer: 7

Let e_k be the expected number of further moves when the token is at k, for $-3 \le k \le 3$, so $e_{-3} = e_3 = 0$. Conditioning on the next move gives, for $-2 \le k \le 2$,

$$e_k = 1 + \tfrac{2}{3}e_{k+1} + \tfrac{1}{3}e_{k-1}.$$

Multiplying by 3 and rearranging, $2(e_k - e_{k+1}) = 3 + (e_{k-1} - e_k)$. So if we write $d_k = e_{k-1} - e_k$ for $-2 \le k \le 3$, then $d_{k+1} = \frac{3+d_k}{2}$.

Let $t = d_{-2} = -e_{-2}$. Then $d_{-1} = \frac{3+t}{2}$, $d_0 = \frac{9+t}{4}$, $d_1 = \frac{21+t}{8}$, $d_2 = \frac{45+t}{16}$ and $d_3 = \frac{93+t}{32}$. These six differences add up to $e_{-3} - e_3 = 0$. Multiplying the sum by 32 gives $63t + 387 = 0$, so $t = -\frac{43}{7}$. Finally

$$e_0 = -(d_{-2} + d_{-1} + d_0) = -\frac{7t + 15}{4} = -\frac{-43 + 15}{4} = 7.$$

Here is a check. A similar calculation with probabilities shows that the token finishes at 3 with probability $\frac{8}{9}$, so its expected final position is $3 \cdot \frac{8}{9} - 3 \cdot \frac{1}{9} = \frac{7}{3}$. Each move adds $\frac{2}{3} - \frac{1}{3} = \frac{1}{3}$ to the position on average, so the expected final position should be $\frac{1}{3}$ of the expected number of moves, and indeed $\frac{1}{3} \cdot 7 = \frac{7}{3}$.

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